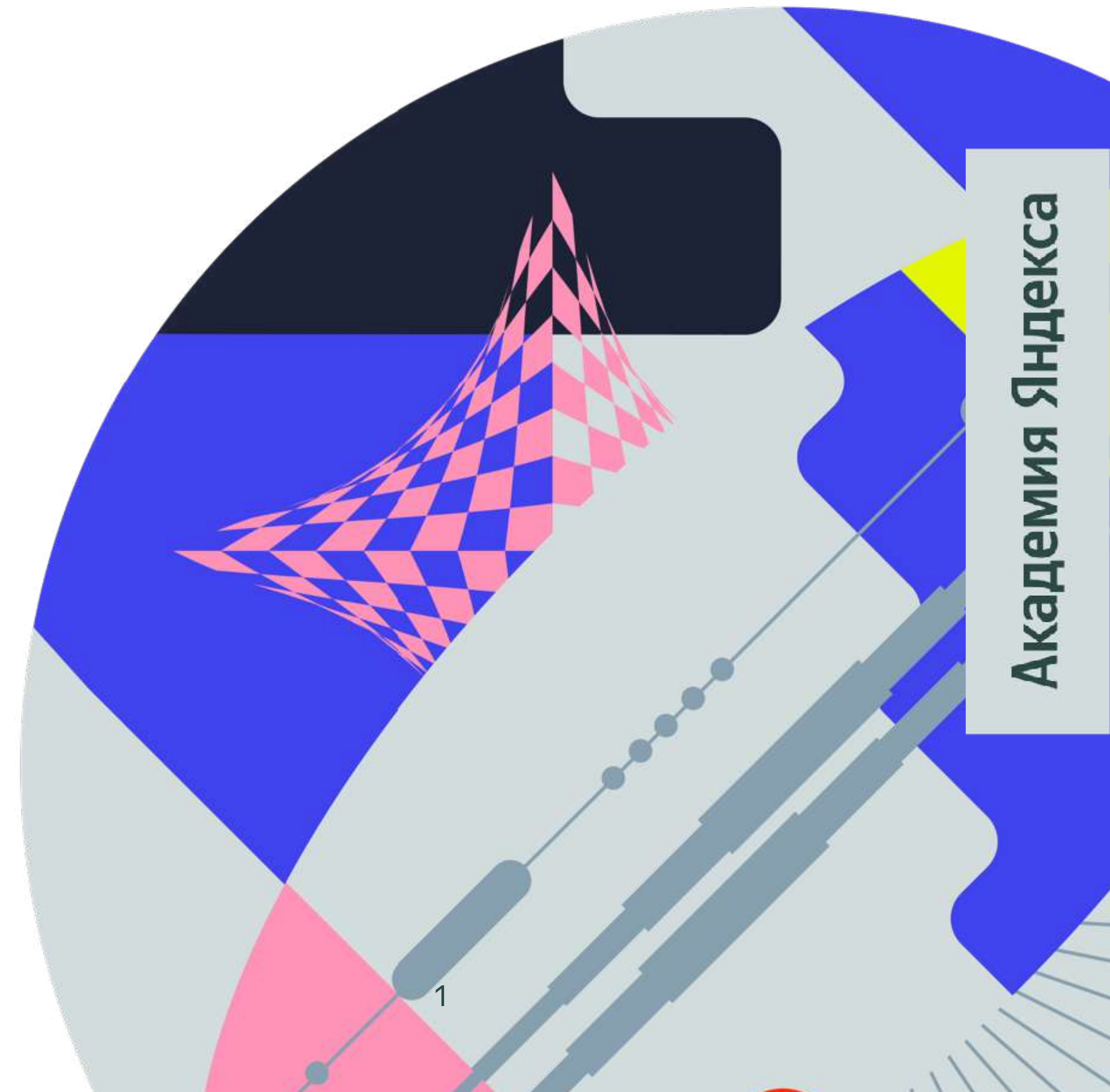


Few-step Generative Models

Nikita Starodubcev

Yandex Research



Lecture 5 | Few-step Generative Models

Part 1. Motivation. Why Few-Step Models?

- Why DMs are slow?
- Naive solution: step knowledge distillation

Part 2. Traditional view on few-step models

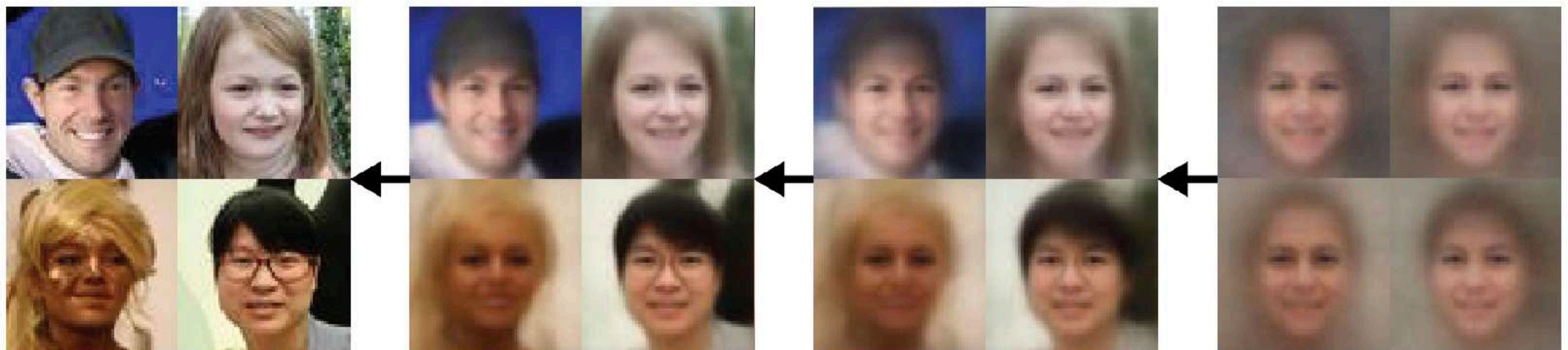
- Consistency Models (CMs)
- Continuous CMs
- Multistep CMs (Multi-boundary)
- Flow Maps
- Mean Flows

Part 3. Unified perspective

- How these methods are connected?

Part 1. Motivation

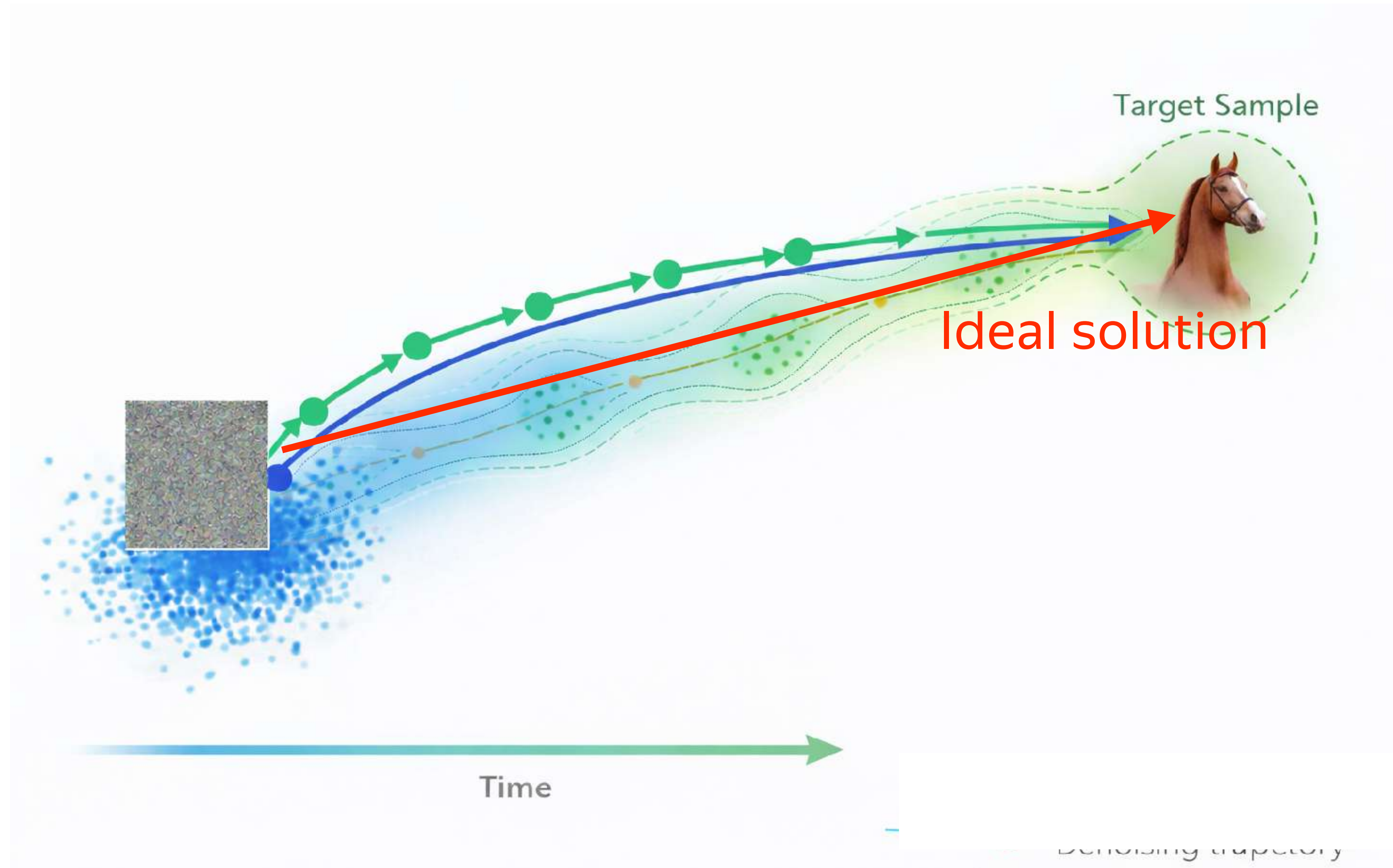
$$\operatorname{argmin}_{\mathbf{x}_\theta} \mathbb{E}_{\mathbf{x}_0, t, \mathbf{x}_t} \|\mathbf{x}_\theta(\mathbf{x}_t, t) - \mathbf{x}_0\|_2^2 = \mathbf{x}_{\theta^*}(\mathbf{x}_t, t) = \frac{\sum_{i=1}^N \alpha_i \mathbf{x}_0^i}{\sum_{i=1}^N \alpha_i}$$
$$\mathbf{x}_{\theta^*}(\mathbf{x}_\sigma, \sigma) = \frac{1}{N} \sum_{i=1}^N \mathbf{x}_0^i$$



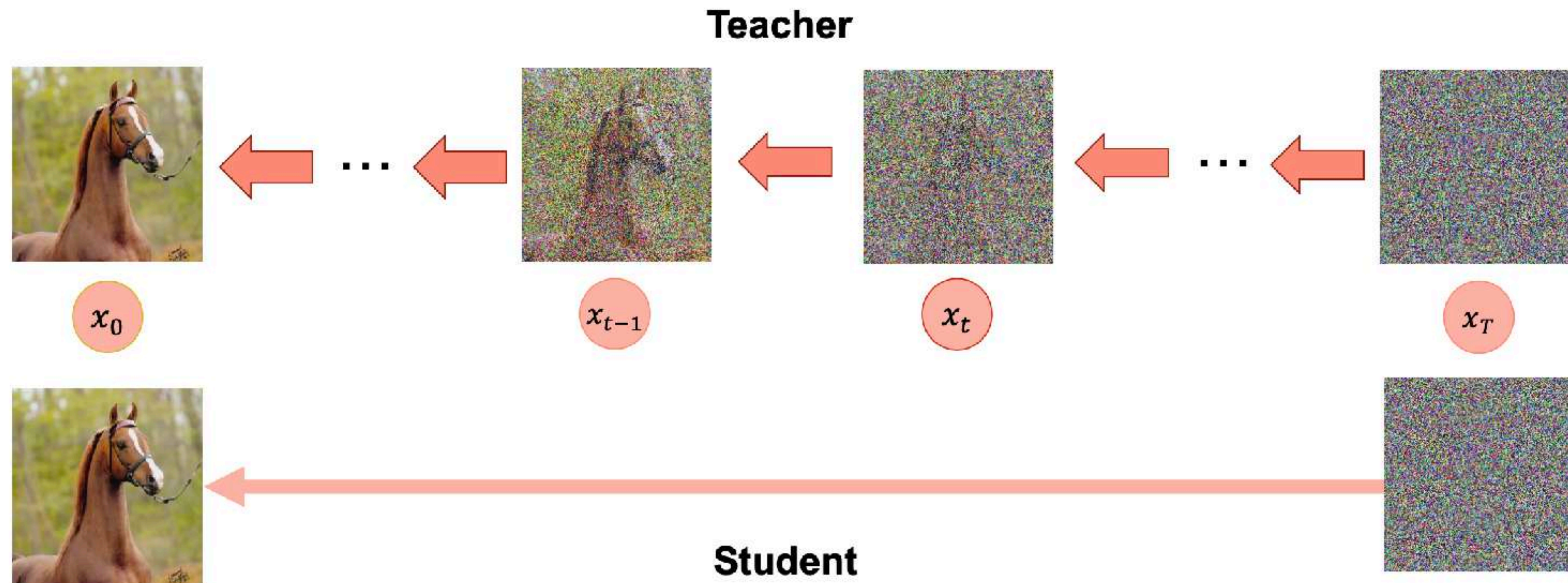
It is impossible to generate samples in a single step
We make images less and less averaged

$$\mathbf{x}_{\theta^*}(\mathbf{x}_t, t) \stackrel{?}{=} \mathbf{x}_0^k$$

DMs have curved trajectories



Knowledge distillation



Main idea

- collect the data using diffusion, $\{(\mathbf{x}_\sigma^1, \mathbf{x}_0^1), \dots, (\mathbf{x}_\sigma^N, \mathbf{x}_0^N)\}$
- fine-tune the network, $\mathbf{x}_\theta(\mathbf{x})$, on this data, $\mathbf{x}_\theta(\mathbf{x}_\sigma) \rightarrow \mathbf{x}_0$

$$\mathbf{L}_{KD}(\mathbf{x}_\theta) = \mathbb{E}_{\mathcal{N}(\mathbf{x}|0,\sigma^2)} \|\mathbf{x}_\theta(\mathbf{x}) - \hat{\mathbf{x}}_\phi(\mathbf{x})\|_2^2$$

$\hat{\mathbf{x}}_\phi(\mathbf{x})$ — sample from the pretrained DM using solver

$$\mathbf{L}_{KD_m}(\mathbf{x}_\theta) = \mathbb{E}_{t \sim \mathcal{U}(0,\sigma)} \mathbb{E}_{p_{data}(\mathbf{x}_0)} \mathbb{E}_{\mathcal{N}(\mathbf{x}|\mathbf{x}_0,t^2)} \|\mathbf{x}_\theta(\mathbf{x}, t) - \hat{\mathbf{x}}_\phi(\mathbf{x}, t)\|_2^2$$

$$\mathbf{L}_{diff}(\mathbf{x}_\phi) = \mathbb{E}_{t \sim \mathcal{U}(0,\sigma)} \mathbb{E}_{p_{data}(\mathbf{x}_0)} \mathbb{E}_{\mathcal{N}(\mathbf{x}|\mathbf{x}_0,t^2)} \|\mathbf{x}_\phi(\mathbf{x}, t) - \mathbf{x}_0\|_2^2$$

Why is diffusion slow but the KD fast?

$$\nabla_{\mathbf{x}_\theta} \mathbf{L}_{KD_m}(\mathbf{x}_\theta) = 0 - ?$$

$$\mathbf{L}_{KD_m}(\mathbf{x}_\theta) = \mathbb{E}_{t \sim \mathcal{U}(0, \sigma)} \mathbb{E}_{p_{data}(\mathbf{x}_0)} \mathbb{E}_{\mathcal{N}(\mathbf{x} | \mathbf{x}_0, t^2)} \|\mathbf{x}_\theta(\mathbf{x}, t) - \hat{\mathbf{x}}_\phi(\mathbf{x}, t)\|_2^2$$

$$= \mathbb{E}_{t \sim \mathcal{U}(0, \sigma)} \int d\mathbf{x}_0 \frac{1}{M} \sum_{j=1}^M \delta(\mathbf{x}_0 - \mathbf{x}_0^j) \left[\int d\mathbf{x} \mathcal{N}(\mathbf{x} | \mathbf{x}_0, t^2) \|\mathbf{x}_\theta(\mathbf{x}, t) - \hat{\mathbf{x}}_\phi(\mathbf{x}, t)\|_2^2 \right]$$

$$= \mathbb{E}_{t \sim \mathcal{U}(0, \sigma)} \int d\mathbf{x} \frac{1}{M} \sum_{j=1}^M \mathcal{N}(\mathbf{x} | \mathbf{x}_0^j, t^2) \|\mathbf{x}_\theta(\mathbf{x}, t) - \hat{\mathbf{x}}_\phi(\mathbf{x}, t)\|_2^2$$

$$\nabla_{\mathbf{x}_\theta} \mathbf{L}_{KD_m}(\mathbf{x}_\theta) = 0 \rightarrow \nabla_{\mathbf{x}_\theta} \frac{1}{M} \sum_{j=1}^M \mathcal{N}(\mathbf{x} | \mathbf{x}_0^j, t^2) \|\mathbf{x}_\theta(\mathbf{x}, t) - \hat{\mathbf{x}}_\phi(\mathbf{x}, t)\|_2^2 = 0$$

$$\nabla_{\mathbf{x}_\theta} \frac{1}{M} \sum_{j=1}^M \mathcal{N}(\mathbf{x} | \mathbf{x}_0^j, t^2) \|\mathbf{x}_\theta(\mathbf{x}, t) - \hat{\mathbf{x}}_\phi(\mathbf{x}, t)\|_2^2 = 0$$

$$\sum_{j=1}^M \mathcal{N}(\mathbf{x} | \mathbf{x}_0^j, t^2) [\mathbf{x}_\theta(\mathbf{x}, t) - \hat{\mathbf{x}}_\phi(\mathbf{x}, t)] = 0$$

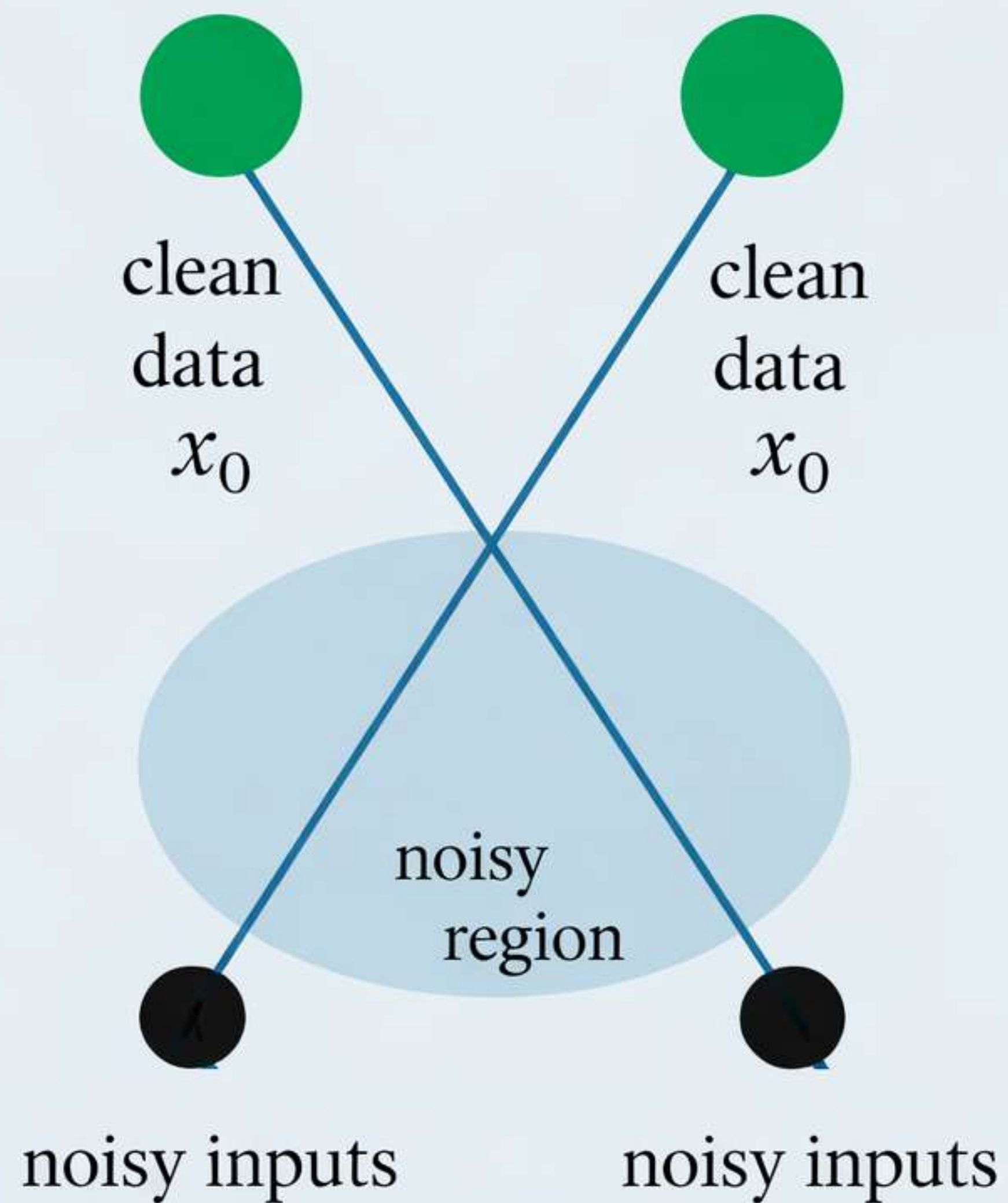
$$\mathbf{x} = \mathbf{x}_0^k + t\mathbf{z} \rightarrow \mathbf{x} = \mathbf{x}(\mathbf{x}_0^k)$$

$$\left[\mathbf{x}_\theta(\mathbf{x}, t) - \hat{\mathbf{x}}_\phi(\mathbf{x}, t) \right] \sum_{j=1}^M \mathcal{N}(\mathbf{x} | \mathbf{x}_0^j, t^2) = 0$$

$$\mathbf{x}_\theta(\mathbf{x}, t) = \hat{\mathbf{x}}_\phi(\mathbf{x}, t)$$

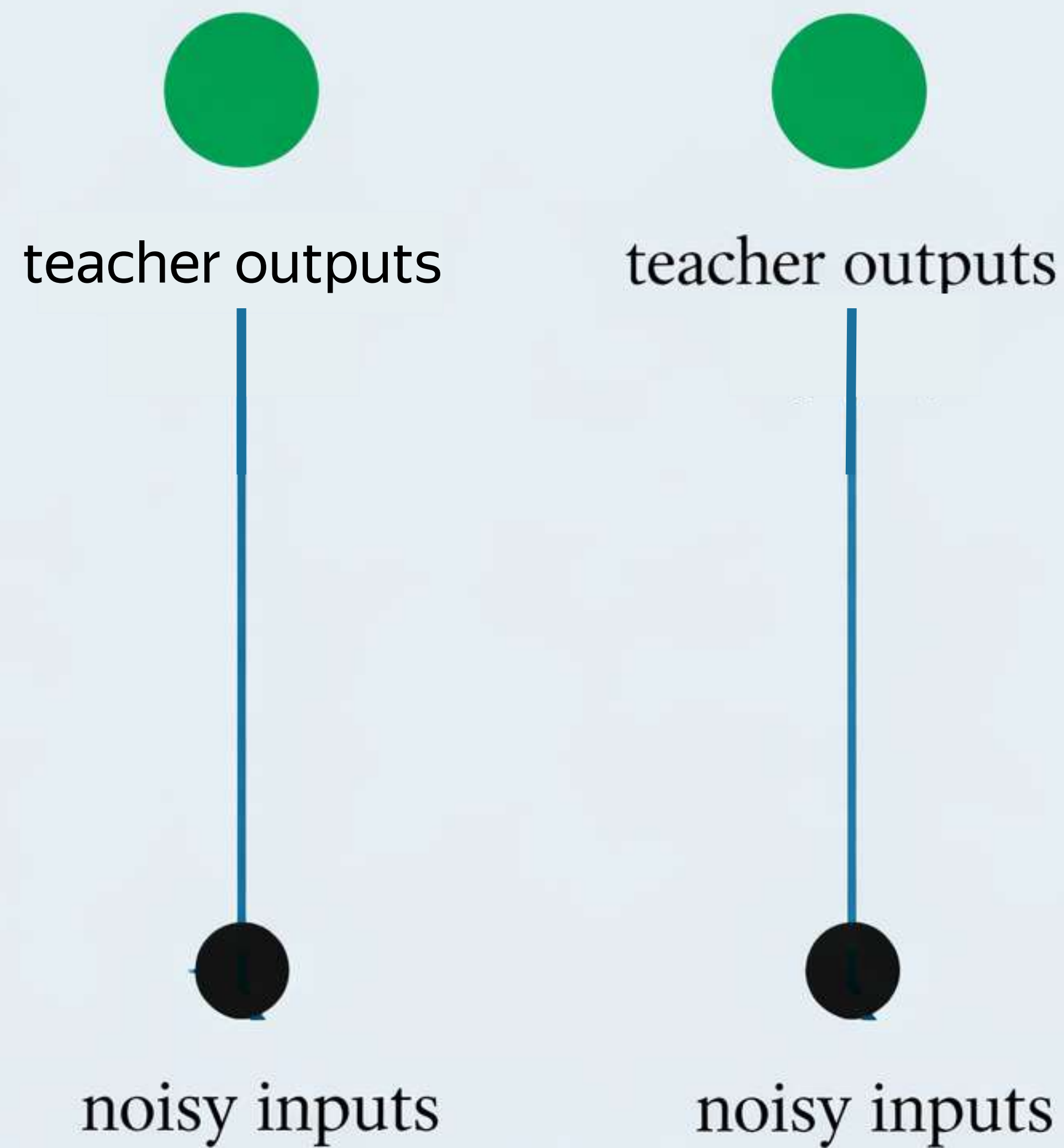
$$\mathbf{x}_\phi(\mathbf{x}, t) = \frac{\sum_{j=1}^N \mathcal{N}(\mathbf{x} | \mathbf{x}_0^j, t^2) \mathbf{x}_0^j}{\sum_{j=1}^N \mathcal{N}(\mathbf{x} | \mathbf{x}_0^j, t^2)}$$

Diffusion training



many-to-one

KD training



one-to-one

General principles

The general principles of constructing ODE-based few-step models

1. Learn a direct mapping along the ODE trajectory instead of its velocity.
2. Use (or do not use) pretrained DM
3. NO GAN LOSS!!!

Part 2. Consistency Models

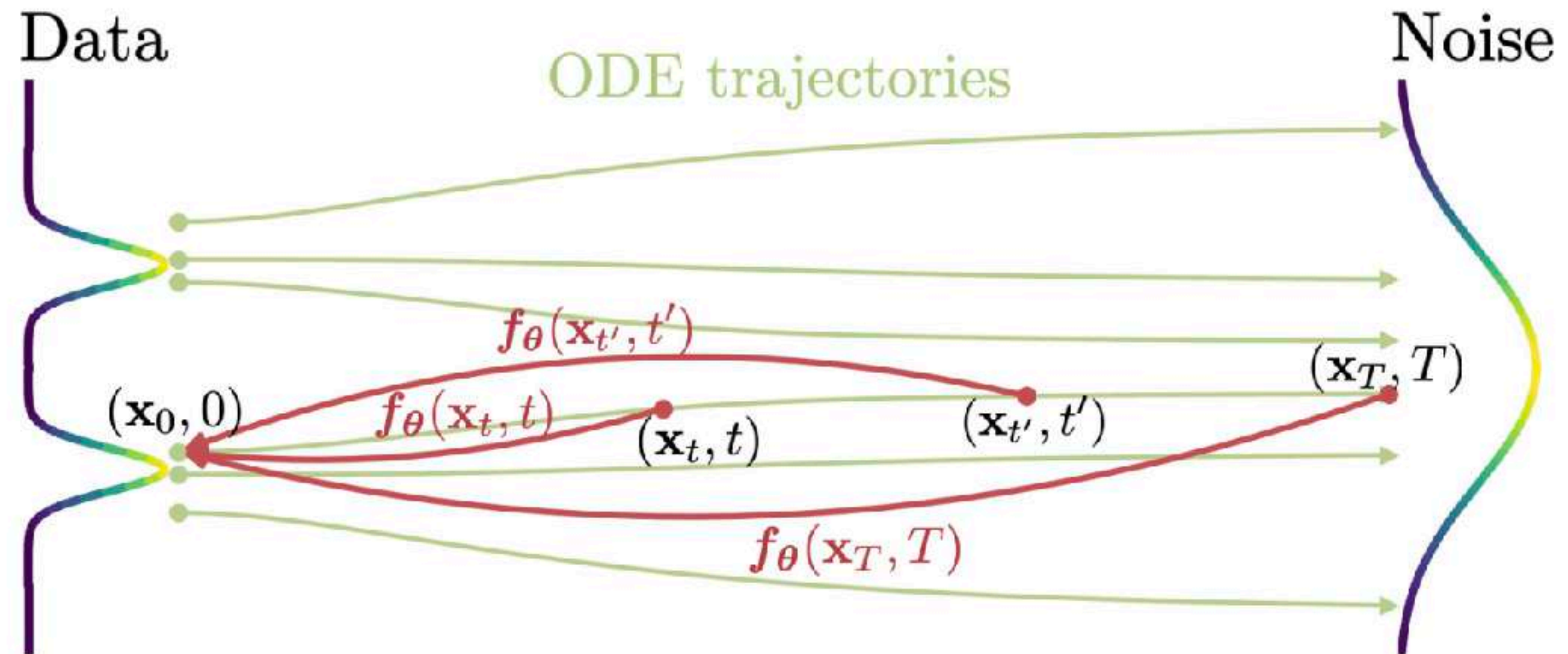
“Learn a direct mapping” What is a mapping?

Definition. Consistency function

Given a solution trajectory of the PF ODE, we define the consistency function as

$$f : (\mathbf{x}_t, t) \rightarrow \mathbf{x}_0, \forall \mathbf{x}_t \in \{\mathbf{x}_\tau\}_{\tau \in [0, \sigma]},$$

where $\{\mathbf{x}_\tau\}_{\tau \in [0, T]}$ points that belong to the same PF ODE trajectory (!)



How to learn a CM?

Self-consistency property

$$f_{\theta}(\mathbf{x}_t, t) = \mathbf{x}_0 \quad \forall \mathbf{x}_t \rightarrow f_{\theta}(\mathbf{x}_0, 0) = \dots = f_{\theta}(\mathbf{x}_s, s) = \dots = f_{\theta}(\mathbf{x}_{\sigma}, \sigma)$$

What if?

Consider discretizing the time horizon $[0, \sigma]$ into N sub-intervals,

$$t_0 = 0 < t_2 < \dots < t_N = \sigma$$

$$\mathbb{E} \left\| f_{\theta}(\mathbf{x}_{n-1}, t_{n-1}) - f_{\theta}(\mathbf{x}_n, t_n) \right\|_2 \rightarrow \min_{\theta}$$

$$\mathbb{E} \left\| f_{\theta}(\mathbf{x}_{n-1}, t_{n-1}) - f_{\theta}(\mathbf{x}_n, t_n) \right\|_2$$

Any issues?

1. We might have a degenerate solution. For example, $f_{\theta}(\mathbf{x}_t, t) = 0$
2. How to get $\{x_n\}$ from the ODE solution?

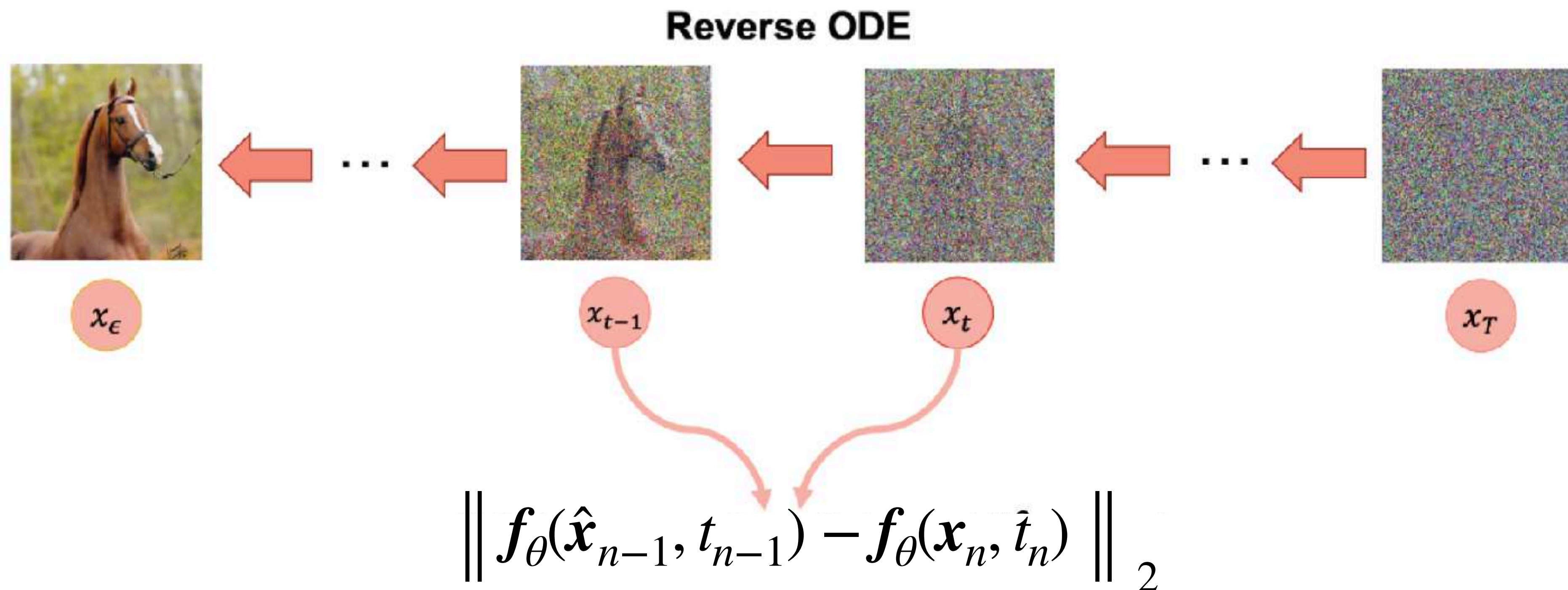
How to fix?

1. Define a **boundary condition** $f_{\theta}(\mathbf{x}_0, t_0) = \mathbf{x}_0$
2. Use a teacher model: $\mathbf{x}_n \sim \mathcal{N}(\mathbf{x}_n | 0, t_n^2)$; $\hat{\mathbf{x}}_{n-1} = \Phi(\mathbf{x}_n, t_n, t_{n-1} | \phi)$

Consistency Distillation. Summary

$$\mathcal{L}_{CD} = \mathbb{E}_{t_n \sim \mathcal{U}(t_0, \dots, t_N)} \mathbb{E}_{\mathbf{x}_0 \sim p_{data}(\mathbf{x}_0)} \mathbb{E}_{\mathbf{x}_n \sim \mathcal{N}(\mathbf{x}_n | \mathbf{x}_0, t_n^2)} \left\| f_{\theta}(\hat{\mathbf{x}}_{n-1}, t_{n-1}) - f_{\theta}(\mathbf{x}_n, t_n) \right\|_2$$

$$\hat{\mathbf{x}}_{n-1} = \Phi(\mathbf{x}_n, t_n, t_{n-1} | \phi), f_{\theta}(\mathbf{x}_0, t_0) = \mathbf{x}_0$$



Consistency Training

$$\left\| f_{\theta}(\mathbf{x}_{n-1}, t_{n-1}) - f_{\theta}(\mathbf{x}_n, t_n) \right\|_2$$

What if we no longer have a teacher?

We still do $\mathbf{x}_n \sim \mathcal{N}(\mathbf{x}_n | 0, t_n^2)$. But how to get $\mathbf{x}_{n-1}$ now?

$$d\mathbf{x} = -t \nabla_{\mathbf{x}} \log p_t(\mathbf{x}) dt$$

$$\nabla_{\mathbf{x}} \log p_t(\mathbf{x}) = -\frac{1}{t^2} \left(\mathbf{x} - \mathbb{E} [\mathbf{x}_0 | \mathbf{x}] \right)$$

Can we estimate this?

$$d\mathbf{x} = -t \nabla_{\mathbf{x}} \log p_t(\mathbf{x}) dt$$

$$\nabla_{\mathbf{x}} \log p_t(\mathbf{x}) = -\frac{1}{t^2} \left(\mathbf{x} - \mathbb{E} [\mathbf{x}_0 | \mathbf{x}] \right) \approx -\frac{1}{t^2} (\mathbf{x} - \mathbf{x}_0)$$

$$d\mathbf{x} = \frac{1}{t} (\mathbf{x} - \mathbf{x}_0) dt$$

$$\int_{t_n}^{t_{n-1}} \frac{d(\mathbf{x} - \mathbf{x}_0)}{(\mathbf{x} - \mathbf{x}_0)} = \frac{dt}{t}$$

$$x_{n-1} = x_0 + \frac{t_{n-1}}{t_n} (x_n - x_0) = x_0 + t_{n-1} \epsilon$$

$$\mathcal{L}_{CT} = \mathbb{E}_{t_n \sim \mathcal{U}(t_0, \dots, t_N)} \mathbb{E}_{\mathbf{x}_0 \sim p_{data}(\mathbf{x}_0)} \mathbb{E}_{x_n \sim \mathcal{N}(\mathbf{x}_n | \mathbf{x}_0, t_n^2)} \left\| f_{\theta}(\mathbf{x}_{n-1}, t_{n-1}) - f_{\theta}(\mathbf{x}_n, t_n) \right\|_2$$

$$\mathbf{x}_n = \mathbf{x}_0 + t_n \epsilon \qquad \mathbf{x}_{n-1} = \mathbf{x}_0 + t_{n-1} \epsilon$$

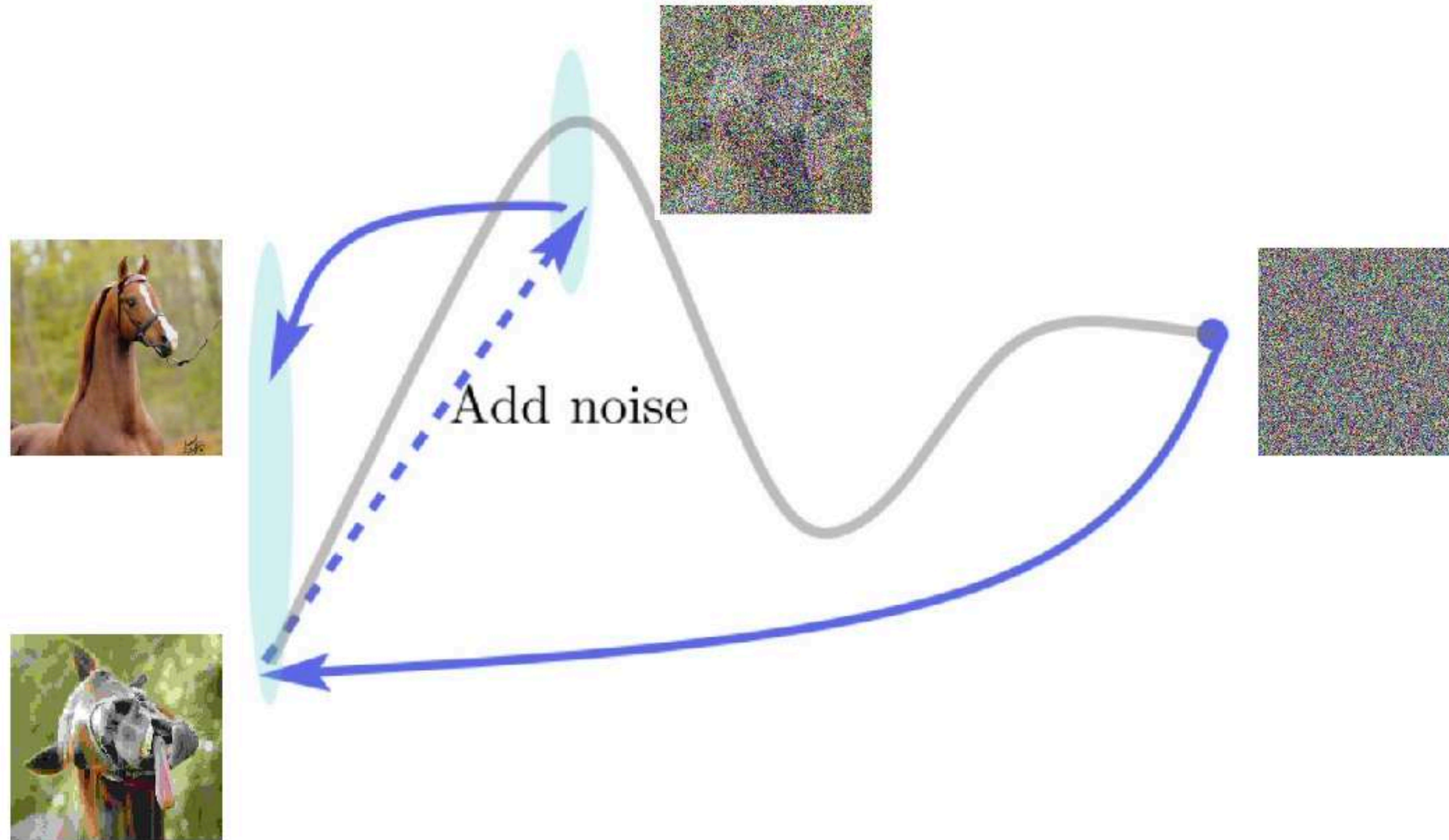
Sampling from CMs

Can we apply ODE solvers to CMs?

$f_{\theta}(\mathbf{x}_n, t_n)$ is not a score function anymore $\rightarrow$ **we cannot use solvers to generate data**

Sampling from CMs

Stochastic multistep sampling



$$\mathbf{x}_0 = f_{\theta}(\mathbf{x}_N, \sigma), \mathbf{x}_N \sim \mathcal{N}(0, \sigma^2)$$

for $n = N - 1, \dots, 1$ **do**

Sample $\epsilon \sim \mathcal{N}(0, 1)$

$$\mathbf{x}_n = \mathbf{x}_0 + t_n \epsilon$$

$$\mathbf{x}_0 = f_{\theta}(\mathbf{x}_n, t_n)$$

$$0 = t_0 < t_1 < \dots < t_N = \sigma$$

We need about 4-5 steps

Continuous CMs

What is the difference between continuous and discrete DMs? (in terms of time)

Inherently, there is no difference.

Discrete DMs sample discrete time steps;
Continuous DMs sample continuous time.

What about CMs?

$$\mathbb{E} \left\| f_{\theta}(\mathbf{x}_{n-1}, t_{n-1}) - f_{\theta}(\mathbf{x}_n, t_n) \right\|_2$$

The task is discrete in terms of time, $t_0 = 0 < t_2 < \dots < t_N = \sigma$

$$\mathbb{E} \left\| f_{\theta}(\mathbf{x}_{n-1}, t_{n-1}) - f_{\theta}(\mathbf{x}_n, t_n) \right\|_2$$

The task is discrete in terms of time.

The main obstacles is that **we need dynamic time discretization.**

“We propose to progressively increase N during training according to a schedule function. The consistency training loss has less “variance” but more “bias” when N is small, which facilitates faster convergence at the beginning of training. On the contrary, it has more “variance” but less “bias” when N is large (i.e., Δt is small), which is desirable when closer to the end of training.”

Statement. Continuous CMs

$$\lim_{\Delta t_n \rightarrow 0} \frac{1}{\Delta t_n} \nabla_{\theta} \mathcal{L}_{CM}^{\Delta t_n} = \nabla_{\theta} \mathcal{L}^{\text{inf}}$$

$$\mathcal{L}^{\text{inf}} = \mathbb{E} \left[f_{\theta}(\mathbf{x}_t, t) \frac{df_{\theta}(\mathbf{x}_t, t)}{dt} \right]$$

$$\frac{df_{\theta}(\mathbf{x}_t, t)}{dt} = \frac{\partial f_{\theta}(\mathbf{x}_t, t)}{\partial t} + \frac{\partial f_{\theta}(\mathbf{x}_t, t)}{\partial \mathbf{x}_t} \frac{d\mathbf{x}_t}{dt}$$

1. What is the difference between CT and CD in this case?
2. $\mathcal{L}^{\text{inf}}$ makes sense only in terms of gradient (pseudo-loss)
 1. The loss does not show the convergence
 2. **We can define different $\mathcal{L}^{\text{inf}}$**
3. We need JVP

Summary for discrete CMs / cont. CMs

Discrete CMs		Continuous CMs	
$\mathbb{E}_{t_n \sim \mathcal{U}(t_0, \dots, t_N)} \mathbb{E}_{\mathbf{x}_0 \sim p_{data}(\mathbf{x}_0)} \mathbb{E}_{\mathbf{x}_n \sim \mathcal{N}(\mathbf{x}_n \mathbf{x}_0, t_n^2)} \left\ \mathbf{f}_\theta(\hat{\mathbf{x}}_{n-1}, t_{n-1}) - \mathbf{f}_\theta(\mathbf{x}_n, t_n) \right\ ^2$		$\mathbb{E}_{t \sim \mathcal{U}(0,1)} \mathbb{E}_{\mathbf{x}_0 \sim p_{data}(\mathbf{x}_0)} \mathbb{E}_{\mathbf{x}_t \sim \mathcal{N}(\mathbf{x} \mathbf{x}_0, t^2)} \left[\mathbf{f}_\theta(\mathbf{x}_t, t) \frac{d\mathbf{f}_\theta(\mathbf{x}_t, t)}{dt} \right]$	
$\mathbf{f}_\theta(\mathbf{x}_0, t_0) = \mathbf{x}_0$		$\frac{d\mathbf{f}_\theta(\mathbf{x}_t, t)}{dt} = \frac{\partial \mathbf{f}_\theta(\mathbf{x}_t, t)}{\partial t} + \frac{\partial \mathbf{f}_\theta(\mathbf{x}_t, t)}{\partial \mathbf{x}_t} \frac{d\mathbf{x}_t}{dt}$	
Distillation	Training	Distillation	Training
$\mathbf{x}_n = \mathbf{x}_0 + t_n \epsilon$	$\mathbf{x}_n = \mathbf{x}_0 + t_n \epsilon$	$\mathbf{x}_n = \mathbf{x}_0 + t_n \epsilon$	$\mathbf{x}_n = \mathbf{x}_0 + t_n \epsilon$
$\hat{\mathbf{x}}_{n-1} = \Phi(\mathbf{x}_n, t_n, t_{n-1} \phi)$	$\hat{\mathbf{x}}_{n-1} = \mathbf{x}_0 + t_{n-1} \epsilon$	$\frac{d\mathbf{x}_t}{dt} = -t s_\phi(\mathbf{x}_t, t)$	$\frac{d\mathbf{x}_t}{dt} = \frac{\mathbf{x}_t - \mathbf{x}_0}{t}$
$\mathbf{f}_\theta = \text{Init}(\mathbf{x}_\phi)$	$\mathbf{f}_\theta = \text{Init}(0)$	$\mathbf{f}_\theta = \text{Init}(\mathbf{x}_\phi)$	$\mathbf{f}_\theta = \text{Init}(0)$
Sampling			
$\mathbf{x}_0 = \mathbf{f}_\theta(\mathbf{x}_N, \sigma), \mathbf{x}_N \sim \mathcal{N}(0, \sigma^2)$		$0 = t_0 < t_1 < \dots < t_N = \sigma$	
for $n = N - 1, \dots, 1$ do		Visit the sampling noise levels	
Sample $\epsilon \sim \mathcal{N}(0,1)$		in decreasing order.	
$\mathbf{x}_n = \mathbf{x}_0 + t_n \epsilon$			
$\mathbf{x}_0 = \mathbf{f}_\theta(\mathbf{x}_n, t_n)$			

Multi-boundary (Multistep) CMs (Consistency Trajectory Models)

The core issue is that we need to approximate the entire solution interval

Can we solve that?

Multi-boundary (Multistep) CMs

The core issue is that we need to approximate the entire solution interval

Can we solve that?

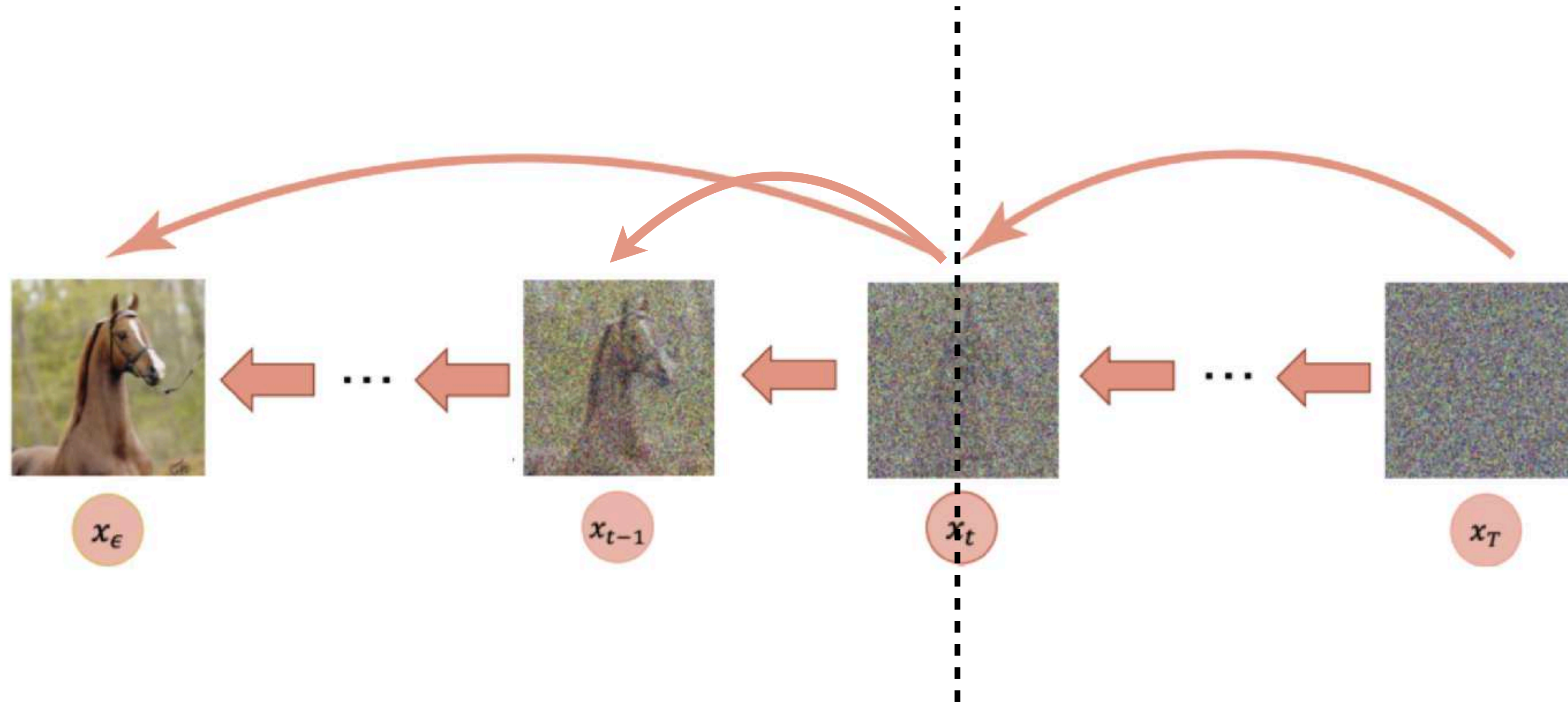
Definition. Multistep (Multi-boundary) Consistency function

Given a solution trajectory of the PF ODE **on the interval** $[t, s]$,
we define the **multistep consistency function** as

$$f : (x_r, r, s) \rightarrow x_s, \forall x_r \in \{x_\tau\}_{\tau \in [t, s]},$$

where $\{x_\tau\}_{\tau \in [t, s]}$ points that belong to the same PF ODE trajectory **on the interval** $[t, s]$.

Multi-boundary (Multistep) CMs



How to learn a multistep CMs?

1. Split the time horizon $[0, \sigma]$ into N intervals, $[t_0, t_s], \dots, [t_m, \sigma]$

N -parts of self-consistency properties:

$$[t_0, t_s] : f_{\theta}(\mathbf{x}_t, t, t_s) = \mathbf{x}_s \quad \forall \mathbf{x}_t \rightarrow f_{\theta}(\mathbf{x}_0, t_0, t_s) = \dots = f_{\theta}(\mathbf{x}_s, t_s, t_s)$$

...

$$[t_m, \sigma] : f_{\theta}(\mathbf{x}_t, t, \sigma) = \mathbf{x}_m \quad \forall \mathbf{x}_t \rightarrow f_{\theta}(\mathbf{x}_m, t_m, \sigma) = \dots = f_{\theta}(\mathbf{x}_{\sigma}, \sigma, \sigma)$$

2. Learn a CM on each interval

$$\mathbb{E}_{t_n \sim \mathcal{U}(t_0, \dots, t_N)} \mathbb{E}_{\mathbf{x}_0 \sim p_{data}(\mathbf{x}_0)} \mathbb{E}_{\mathbf{x}_n \sim \mathcal{N}(\mathbf{x}_n | \mathbf{x}_0, t_n^2)} \left\| f_{\theta}(\mathbf{x}_{n-1}, t_{n-1}, t'_n) - f_{\theta}(\mathbf{x}_n, t_n, t'_n) \right\|_2$$

$$t'_n = F_{boundary}(t_n)$$

Ours, 4 steps



Teacher, 50 steps



*A donkey in a clown costume giving a lecture at the front of a lecture hall.
The blackboard has mathematical equations on it.*

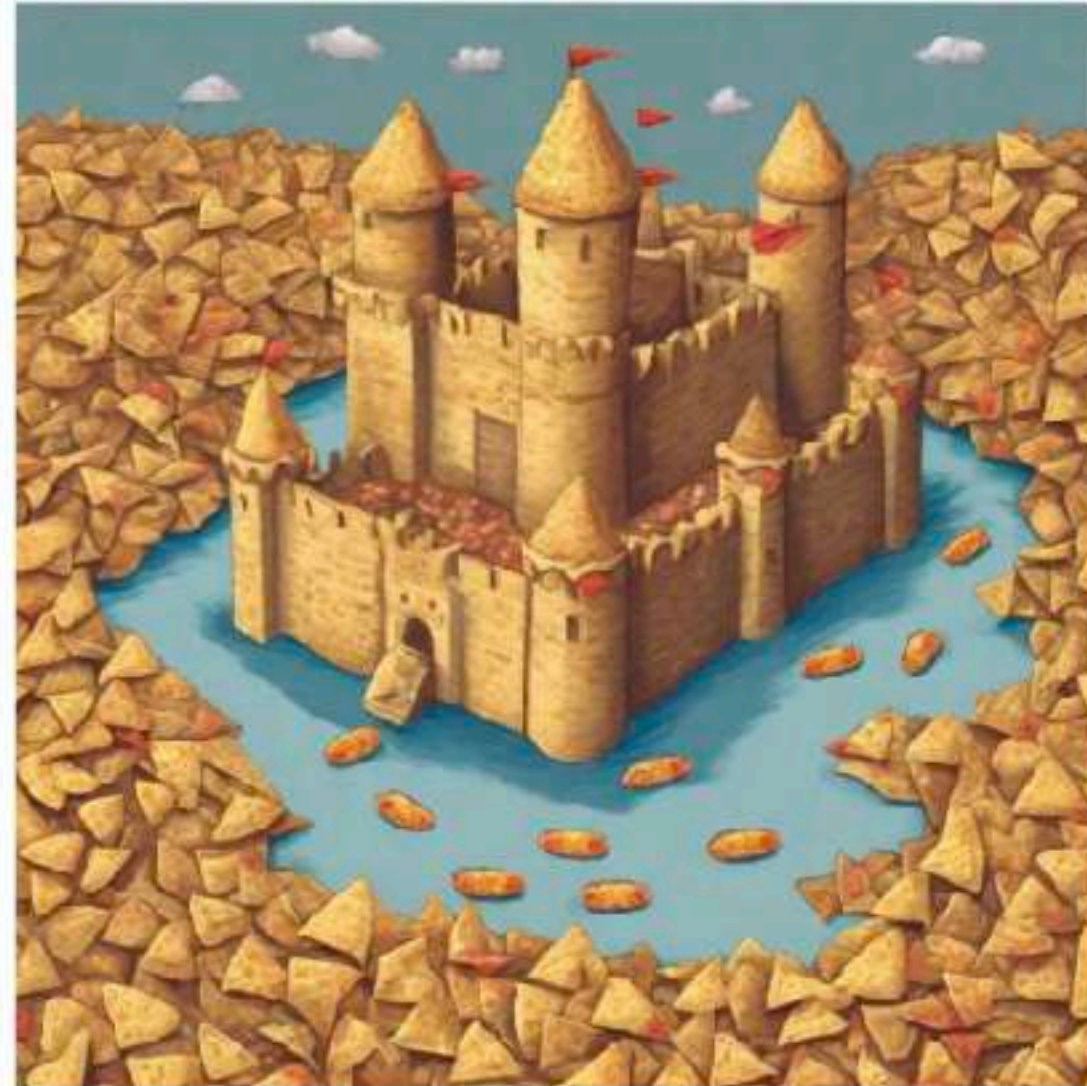
Ours, 4 steps



Teacher, 50 steps



an airplane taxiing on a runway with the sun behind it



*A castle made of tortilla chips, in a river made of salsa.
There are tiny burritos walking around the castle*



*A raccoon wearing formal clothes, wearing a top hat and holding a cane.
The raccoon is holding a garbage bag. Oil painting in the style of cubism.*

Continuous multistep CMs

“Split the time horizon $[0, \sigma]$ into N intervals, $[t_0, t_s], \dots [t_m, \sigma]$ ”

In the continuous case, we have an infinite number of intervals.

That is, the model will learn any transition $t \rightarrow s$

Statement. Continuous multistep CMs (Flow Maps)

$$\mathcal{L}_{\theta}^{FMs} = \text{sign}(s - t) \cdot f_{\theta}^T(z_s, s, t) \frac{df_{\theta}(z_s, s, t)}{ds}$$

Summary for multistep discrete CMs / cont. CMs

Discrete mCMs		Continuos mCMs	
$\mathbb{E}_{t_n \sim \mathcal{U}(t_0, \dots, t_N)} \mathbb{E}_{\mathbf{x}_0 \sim p_{data}(\mathbf{x}_0)} \mathbb{E}_{\mathbf{x}_n \sim \mathcal{N}(\mathbf{x}_n \mathbf{x}_0, t_n^2)} \left\ \mathbf{f}_\theta(\mathbf{x}_{n-1}, t_{n-1}, t'_n) - \mathbf{f}_\theta(\mathbf{x}_n, t_n, t'_n) \right\ ^2$		$\mathcal{L}_\theta^{FMs} = sign(s - t) \mathbf{f}_\theta^T(\mathbf{x}_s, s, t) \frac{d\mathbf{f}_\theta(\mathbf{x}_s, s, t)}{ds}$	
$t'_n = F_{boundary}(t_n), \mathbf{f}_\theta(\mathbf{x}_n, t_n, t'_n) = \mathbf{x}_n$		$\frac{d\mathbf{f}_\theta(\mathbf{x}_s, s, t)}{ds} = \frac{\partial \mathbf{f}_\theta(\mathbf{x}_s, s, t)}{\partial s} + \frac{\partial \mathbf{f}_\theta(\mathbf{x}_s, s, t)}{\partial \mathbf{x}_s} \frac{d\mathbf{x}_s}{ds}$	
Distillation	Training	Distillation	Training
$\mathbf{x}_n = \mathbf{x}_0 + t_n \epsilon$	$\mathbf{x}_n = \mathbf{x}_0 + t_n \epsilon$	$\mathbf{x}_n = \mathbf{x}_0 + t_n \epsilon$	$\mathbf{x}_n = \mathbf{x}_0 + t_n \epsilon$
$\hat{\mathbf{x}}_{n-1} = \Phi(\mathbf{x}_n, t_n, t_{n-1} \phi)$	$\hat{\mathbf{x}}_{n-1} = \mathbf{x}_0 + t_{n-1} \epsilon$	$\frac{d\mathbf{x}_t}{dt} = -ts_\phi(\mathbf{x}_t, t)$	$\frac{d\mathbf{x}_t}{dt} = \frac{\mathbf{x}_t - \mathbf{x}_0}{t}$
$\mathbf{f}_\theta = \text{Init}(\mathbf{x}_\phi)$	$\mathbf{f}_\theta = \text{Init}(0)$	$\mathbf{f}_\theta = \text{Init}(\mathbf{x}_\phi)$	$\mathbf{f}_\theta = \text{Init}(0)$
Sampling			
$\mathbf{x} = \mathbf{f}_\theta(\mathbf{x}_N, \sigma), \mathbf{x}_N \sim \mathcal{N}(0, \sigma^2)$			
for $n = 1$ to $N - 1$ do			
$\mathbf{x} = \mathbf{f}_\theta(\mathbf{x}, t_n)$			

MeanFlows, Shortcut (in short)

After that, a huge amount of work came out...

Stable consistency tuning: Understanding and improving consistency models

[FY Wang](#), [Z Geng](#), H Li - arXiv preprint arXiv:2410.18958, 2024 - arxiv.org

Diffusion models achieve superior generation quality but suffer from slow generation speed due to the iterative nature of denoising. In contrast, consistency models, a new generative ...

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One-step diffusion distillation via deep equilibrium models

[Z Geng](#), [A POKLE](#), [JZ Kolter](#) - Advances in Neural ..., 2023 - proceedings.neurips.cc

Diffusion models excel at producing high-quality samples but naively require hundreds of iterations, prompting multiple attempts to distill the generation process into a faster network ...

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One-step Diffusion Models with f -Divergence Distribution Matching

[Y Xu](#), [W Nie](#), [A Vahdat](#) - arXiv preprint arXiv:2502.15681, 2025 - arxiv.org

Sampling from diffusion models involves a slow iterative process that hinders their practical deployment, especially for interactive applications. To accelerate generation speed, recent ...

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Align your flow: Scaling continuous-time flow map distillation

[A Sabour](#), [S Fidler](#), [K Kreis](#) - arXiv preprint arXiv:2506.14603, 2025 - arxiv.org

Diffusion-and flow-based models have emerged as state-of-the-art generative modeling approaches, but they require many sampling steps. Consistency models can distill these ...

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Consistency models made easy

[Z Geng](#), [A POKLE](#), [W Luo](#), [J Lin](#), [JZ Kolter](#) - arXiv preprint arXiv:2406.14548, 2024 - arxiv.org

Consistency models (CMs) offer faster sampling than traditional diffusion models, but their training is resource-intensive. For example, as of 2024, training a state-of-the-art CM on ...

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One step diffusion via shortcut models

[K Frans](#), [D Hafner](#), [S Levine](#), [P Abbeel](#) - arXiv preprint arXiv:2410.12557, 2024 - arxiv.org

Diffusion models and flow-matching models have enabled generating diverse and realistic images by learning to transfer noise to data. However, sampling from these models involves ...

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Consistency flow matching: Defining straight flows with velocity consistency

[L Yang](#), [Z Zhang](#), [Z Zhang](#), X Liu, [M Xu](#)... - arXiv preprint arXiv ..., 2024 - arxiv.org

Flow matching (FM) is a general framework for defining probability paths via Ordinary Differential Equations (ODEs) to transform between noise and data samples. Recent ...

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One-step Latent-free Image Generation with Pixel Mean Flows

[Y Lu](#), S Lu, [Q Sun](#), H Zhao, [Z Jiang](#), [X Wang](#)... - arXiv preprint arXiv ..., 2026 - arxiv.org

Modern diffusion/flow-based models for image generation typically exhibit two core characteristics:(i) using multi-step sampling, and (ii) operating in a latent space. Recent ...

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Mean flows for one-step generative modeling

[Z Geng](#), [M Deng](#), [X Bai](#), [JZ Kolter](#), [K He](#) - arXiv preprint arXiv:2505.13447, 2025 - arxiv.org

We propose a principled and effective framework for one-step generative modeling. We introduce the notion of average velocity to characterize flow fields, in contrast to ...

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Inductive moment matching

[L Zhou](#), [S Ermon](#), [J Song](#) - arXiv preprint arXiv:2503.07565, 2025 - arxiv.org

Diffusion models and Flow Matching generate high-quality samples but are slow at inference, and distilling them into few-step models often leads to instability and extensive ...

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Consistent diffusion models: Mitigating sampling drift by learning to be consistent

[G Daras](#), [Y Dagan](#), [A Dimakis](#)... - Advances in Neural ..., 2023 - proceedings.neurips.cc

Imperfect score-matching leads to a shift between the training and the sampling distribution of diffusion models. Due to the recursive nature of the generation process, errors in previous ...

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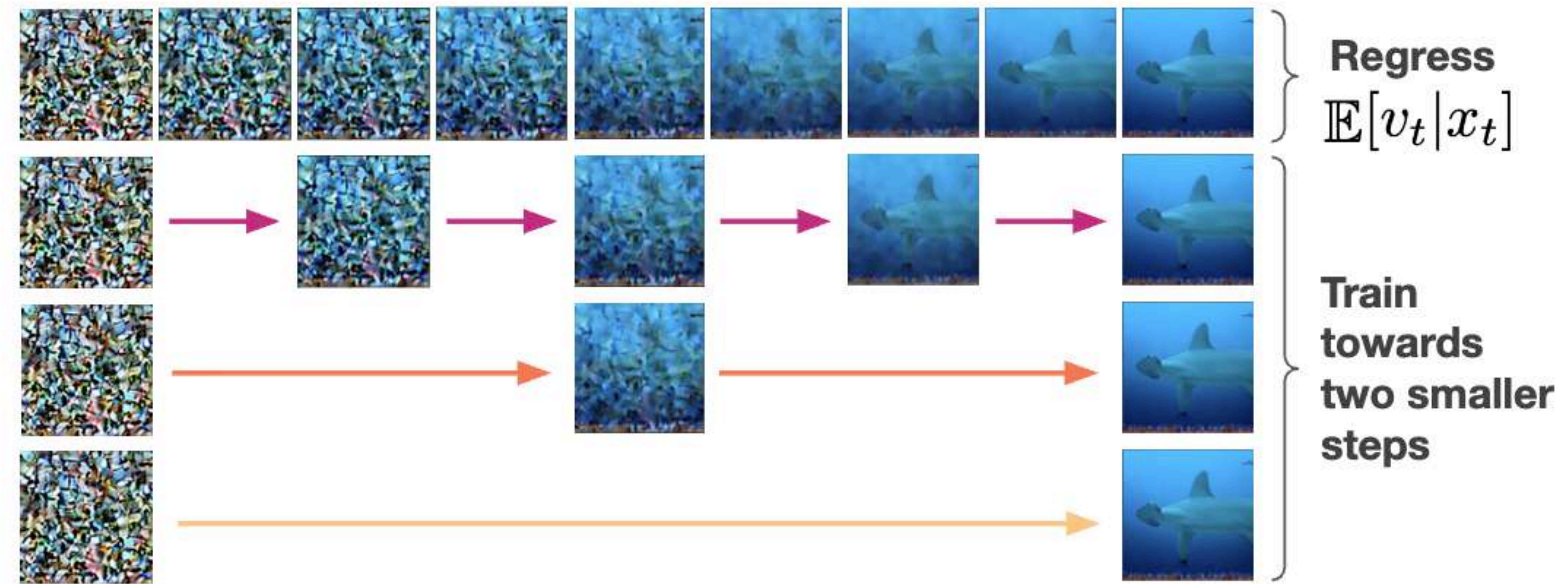
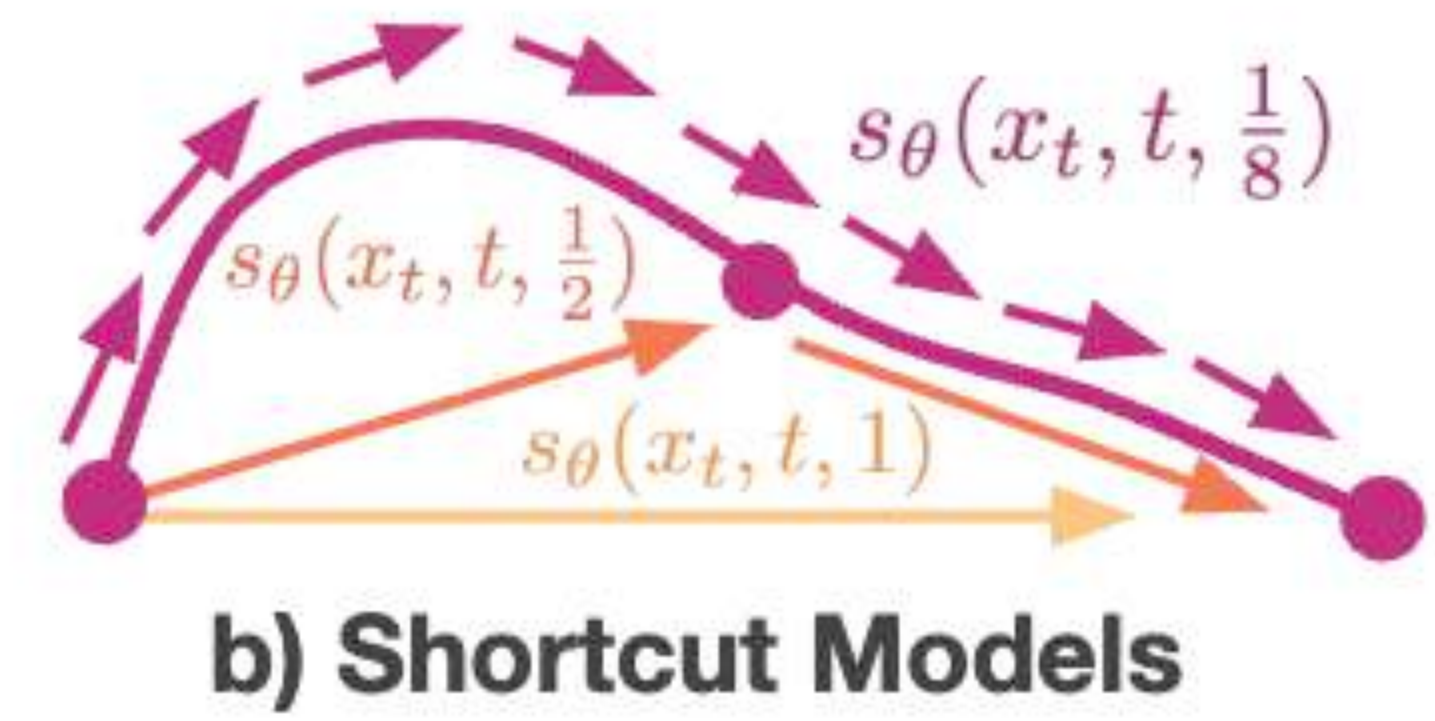
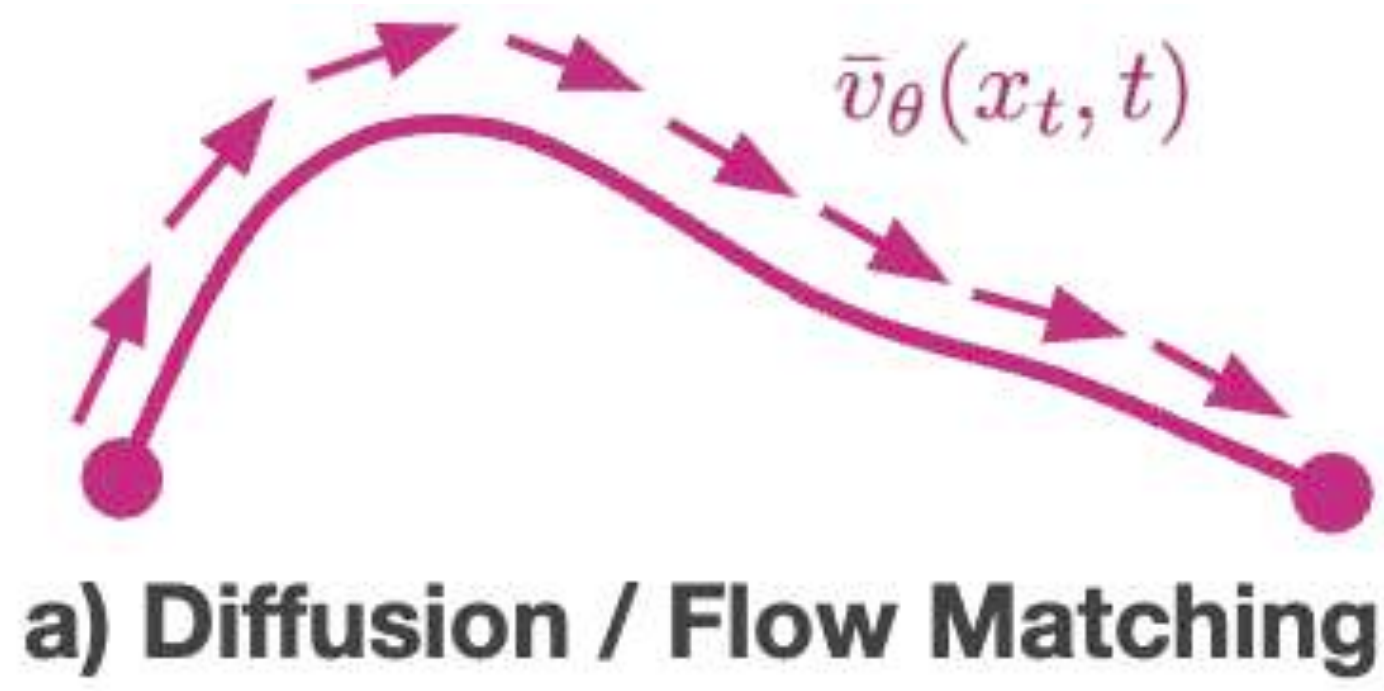
Multistep consistency models

[J Heek](#), [E Hoogetboom](#), [T Salimans](#) - arXiv preprint arXiv:2403.06807, 2024 - arxiv.org

Diffusion models are relatively easy to train but require many steps to generate samples. Consistency models are far more difficult to train, but generate samples in a single step. In ...

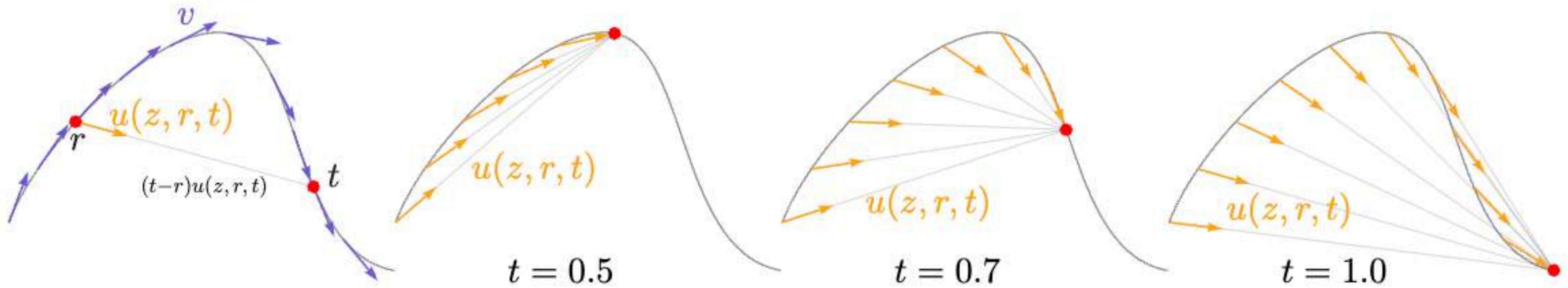
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Shortcut Models



Mean Flows

$$u(z_t, r, t) \triangleq \frac{1}{t-r} \int_r^t v(z_\tau, \tau) d\tau.$$



Part 3. Unified Perspective

So, we studied CMs, Multistep CMs, MeanFlows, Shortcut Models

Lets build a unified perspective to show the connections between them

General principle

Define an "integrator" and say something about its properties

$$\dot{z}_t = v(z_t, t) \rightarrow z_t = z_s + \int_s^t v(z_\tau, \tau) d\tau, t \in [1,0]$$

Define an "integrator" and say something about its properties

Two degrees of freedom: discrete/continuous ; one step/ multistep

1. Discrete, one step

$$f(z_s, s) := z_s + \int_s^0 v(z_\tau, \tau) d\tau$$

Any properties?

$$f(z_s, s) = z_s + \int_s^0 v(z_\tau, \tau) d\tau = z_s + (z_0 - z_s) = z_0$$

$$f(z_s, s) = f(z_r, r)$$

$$f(z_0, 0) = z_0$$

Theorem of calculus

$$\int_a^b g'_\tau d\tau = g(b) - g(a)$$

2. Discrete, multistep

$$f(z_s, s, t) := z_s + \int_s^t v(z_\tau, \tau) d\tau$$

$$f(z_s, s, t) = z_s + \int_s^t v(z_\tau, \tau) d\tau = z_s + (z_t - z_s) = z_t$$

$$f(z_s, s, t) = f(z_r, r, t)$$

$$f(z_t, t, t) = z_t$$

$$\mathcal{L}_\theta^{CTM} = \|f_\theta(z_s, s, t) - f_\theta(z_r, r, t)\|$$

2. Discrete, multistep

$$s(z_s, s, t) := \frac{1}{t - s} \cdot \int_s^t v(z_\tau, \tau) d\tau$$

We can use a different property now

The integral addition property

$$\int_a^b g_\tau d\tau = \int_a^c g_\tau d\tau + \int_c^b g_\tau d\tau$$

$$s(z_s, s, t) = \frac{1}{2} s(z_s, s, r) + \frac{1}{2} s(z_r, r, t), \quad r = \frac{(t + s)}{2}$$

2. Discrete, multistep

$$s(z_s, s, t) = \frac{1}{2}s(z_s, s, r) + \frac{1}{2}s(z_r, r, t), \quad r = \frac{t + s}{2}$$

$$\mathcal{L}_\theta^{SM} = \left\| s_\theta(z_s, s, t) - \frac{1}{2}s_\theta(z_s, s, r) - \frac{1}{2}s_\theta(z_r, r, t) \right\|_2$$

Connection: $\mathcal{L}_\theta^{SM} = \frac{1}{t - s} \mathcal{L}_\theta^{CTM}$

$$\mathcal{L}_\theta^{SM} = \frac{1}{t-s} \mathcal{L}_\theta^{CTM}$$

$$s_\theta(z_s, s, t)(t-s) = f_\theta(z_s, s, t) - z_s$$

$$\mathcal{L}_\theta^{SM} = \left\| s_\theta(z_s, s, t) - \frac{1}{2}s_\theta(z_s, s, r) - \frac{1}{2}s_\theta(z_r, r, t) \right\|_2, \quad r = \frac{t+s}{2}$$

$$\mathcal{L}_\theta^{SM} = \left\| \frac{f_\theta(z_s, s, t) - z_s}{t-s} - \frac{1}{2} \frac{f_\theta(z_s, s, r) - z_s}{r-s} - \frac{1}{2} \frac{f_\theta(z_r, r, t) - z_r}{t-r} \right\|_2$$

$$\mathcal{L}_\theta^{SM} = \left\| \frac{f_\theta(z_s, s, t) - z_s}{t-s} - \frac{f_\theta(z_s, s, r) - z_s}{t-s} - \frac{f_\theta(z_r, r, t) - z_r}{t-s} \right\|_2$$

$$\mathcal{L}_{\theta}^{SM} = \left\| \frac{f_{\theta}(z_s, s, t) - z_s}{t - s} - \frac{f_{\theta}(z_s, s, r) - z_s}{t - s} - \frac{f_{\theta}(z_r, r, t) - z_r}{t - s} \right\|_2$$

$$\mathcal{L}_{\theta}^{SM} = \left\| \frac{f_{\theta}(z_s, s, t)}{t - s} - \frac{f_{\theta}(z_s, s, r)}{t - s} - \frac{f_{\theta}(z_r, r, t) - z_r}{t - s} \right\|_2, \quad z_r = f_{\theta}(z_s, s, r)$$

$$\mathcal{L}_{\theta}^{SM} = \frac{1}{t - s} \left\| f_{\theta}(z_s, s, t) - f_{\theta}(z_r, r, t) \right\|_2 = \frac{1}{t - s} \mathcal{L}_{\theta}^{CTM}$$

3. Continuous, one step

$$f(z_s, s) := z_s + \int_s^0 v(z_\tau, \tau) d\tau$$

$$f(z_s, s) = f(z_r, r)$$

How to write this property in a continuous way?

$$\frac{df(z_s, s)}{ds} = 0$$

$$\mathcal{L}_\theta^{sCM} = \left\| \frac{df(z_s, s)}{ds} \right\|_2$$

4. Continuous, multistep

$$f(z_s, s, t) := z_s + \int_s^t v(z_\tau, \tau) d\tau$$

$$f(z_s, s, t) = f(z_r, r, t)$$

$$\frac{df(z_s, s, t)}{ds} = 0$$

$$\mathcal{L}_\theta^{FlowMaps} = \left\| \frac{df(z_s, s, t)}{ds} \right\|_2$$

4. Continuous, multistep

$$u(z_s, s, t) := \frac{1}{(t - s)} \cdot \int_s^t v(z_\tau, \tau) d\tau$$

$$u(z_s, s, t) = (t - s) \frac{d}{ds} u(z_s, s, t) + v(z_s, s)$$

$$\mathcal{L}_\theta^{MF} = \left\| u_\theta(z_s, s, t) - (t - s) \frac{d}{ds} u_\theta(z_s, s, t) - v(z_s, s) \right\|_2$$

$$u(z_s, s, t) := \frac{1}{t - s} \cdot \int_s^t v(z_\tau, \tau) d\tau$$

a) MeanFlow is a continuous variant of Shortcut models

$$s(z_s, s, t) = \frac{r - s}{t - s} s(z_s, s, r) + \frac{t - r}{t - s} s(z_r, r, t), \quad s < r < t.$$

$$r = s + h, \quad h > 0, \quad h \rightarrow 0.$$

$$s(z_s, s, t) = \frac{h}{t - s} s(z_s, s, s + h) + \frac{t - s - h}{t - s} s(z_{s+h}, s + h, t).$$

$$s(z_s, s, t) = \frac{h}{t-s} s(z_s, s, s+h) + \frac{t-s-h}{t-s} s(z_{s+h}, s+h, t).$$

$$s(z_s, s, s+h) = \frac{1}{h} \int_s^{s+h} v(z_\tau, \tau) \, d\tau.$$

$$s(z_s, s, s+h) \rightarrow v(z_s, s) \quad \text{as } h \rightarrow 0.$$

$$s(z_{s+h}, s+h, t) = s(z_s, s, t) + h \frac{d}{ds} s(z_s, s, t) + O(h^2).$$

$$s(z_s, s, t) = \frac{h}{t-s} v(z_s, s) + \left(1 - \frac{h}{t-s}\right) \left(s(z_s, s, t) + h \frac{d}{ds} s(z_s, s, t) + O(h^2)\right).$$

$$s(z_s, s, t) = \frac{h}{t-s} v(z_s, s) + \left(1 - \frac{h}{t-s}\right) \left(s(z_s, s, t) + h \frac{d}{ds} s(z_s, s, t) + O(h^2)\right).$$

$$s(z_s, s, t) = \frac{h}{t-s} v(z_s, s) + s(z_s, s, t) + h \frac{d}{ds} s(z_s, s, t) - \frac{h}{t-s} s(z_s, s, t) + O(h^2).$$

$$0 = \frac{h}{t-s} v(z_s, s) + h \frac{d}{ds} s(z_s, s, t) - \frac{h}{t-s} s(z_s, s, t) + O(h^2).$$

$$0 = \frac{1}{t-s} v(z_s, s) + \frac{d}{ds} s(z_s, s, t) - \frac{1}{t-s} s(z_s, s, t) + \frac{O(h^2)}{h}.$$

$$0 = \frac{1}{t-s}v(z_s, s) + \frac{d}{ds}s(z_s, s, t) - \frac{1}{t-s}s(z_s, s, t) + \frac{O(h^2)}{h}.$$

$$0 = \frac{1}{t-s}v(z_s, s) + \frac{d}{ds}s(z_s, s, t) - \frac{1}{t-s}s(z_s, s, t).$$

$$s(z_s, s, t) = (t-s)\frac{d}{ds}s(z_s, s, t) + v(z_s, s).$$

$$u(z_s, s, t) := \frac{1}{t - s} \cdot \int_s^t v(z_\tau, \tau) d\tau$$

b) MeanFlow is a continuous version of CTM, but in a different parameterization

$$r = s + h, \quad h > 0, \quad h \rightarrow 0.$$

$$\mathcal{L}_\theta^{\text{CTM}} = \left\| f_\theta(z_s, s, t) - f_\theta(z_r, r, t) \right\|_2$$

$$f_\theta(z_s, s, t) = u_\theta(z_s, s, t)(t - s) + z_s,$$

$$\mathcal{L}_\theta^{\text{CTM}} = \left\| u_\theta(z_s, s, t)(t - s) + z_s - u_\theta(z_{s+h}, s + h, t)(t - s - h) - z_{s+h} \right\|_2$$

$$\mathcal{L}_\theta^{\text{CTM}} = \left| u_\theta(z_s, s, t)(t - s) + z_s - u_\theta(z_{s+h}, s + h, t)(t - s - h) - z_{s+h} \right|.$$

$$\mathcal{L}_\theta^{\text{CTM}} = \left| u_\theta(z_s, s, t)(t - s) - u_\theta(z_{s+h}, s + h, t)(t - s) + u_\theta(z_{s+h}, s + h, t)h + z_s - z_{s+h} \right|.$$

$$\frac{1}{h}\mathcal{L}_\theta^{\text{CTM}} = \left| (t - s)\frac{u_\theta(z_s, s, t) - u_\theta(z_{s+h}, s + h, t)}{h} + u_\theta(z_{s+h}, s + h, t) + \frac{z_s - z_{s+h}}{h} \right|.$$

$$\begin{aligned} \frac{u_\theta(z_s, s, t) - u_\theta(z_{s+h}, s + h, t)}{h} &\rightarrow -\frac{d}{ds}u_\theta(z_s, s, t), & u_\theta(z_{s+h}, s + h, t) &\rightarrow u_\theta(z_s, s, t). \\ \frac{z_s - z_{s+h}}{h} &\rightarrow -v(z_s, s). \end{aligned}$$

$$\lim_{h \rightarrow 0^+} \frac{1}{h}\mathcal{L}_\theta^{\text{CTM}} = \left| -(t - s)\frac{d}{ds}u_\theta(z_s, s, t) + u_\theta(z_s, s, t) - v(z_s, s) \right|.$$

$$\lim_{h \rightarrow 0^+} \frac{1}{h} \mathcal{L}_\theta^{\text{CTM}} = \left| -(t-s) \frac{d}{ds} u_\theta(z_s, s, t) + u_\theta(z_s, s, t) - v(z_s, s) \right|.$$

$$\mathcal{L}_\theta^{\text{MF}} := \lim_{h \rightarrow 0^+} \frac{1}{h} \mathcal{L}_\theta^{\text{CTM}} = \left| u_\theta(z_s, s, t) - (t-s) \frac{d}{ds} u_\theta(z_s, s, t) - v(z_s, s) \right|.$$

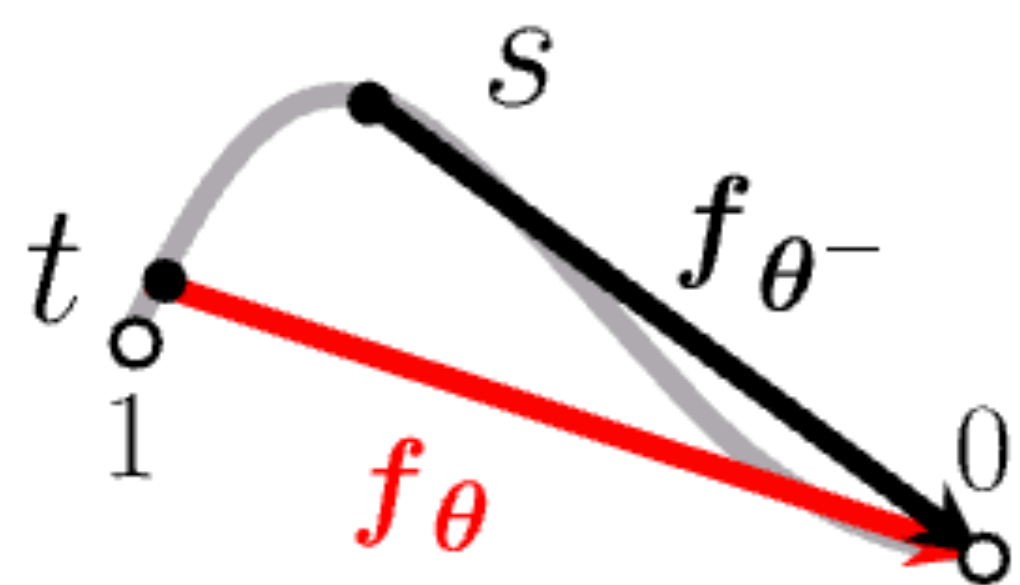
$$u(z_s, s, t) := \frac{1}{t - s} \cdot \int_s^t v(z_\tau, \tau) d\tau$$

c) MeanFlow is equivalent to Flow Maps in terms of gradients

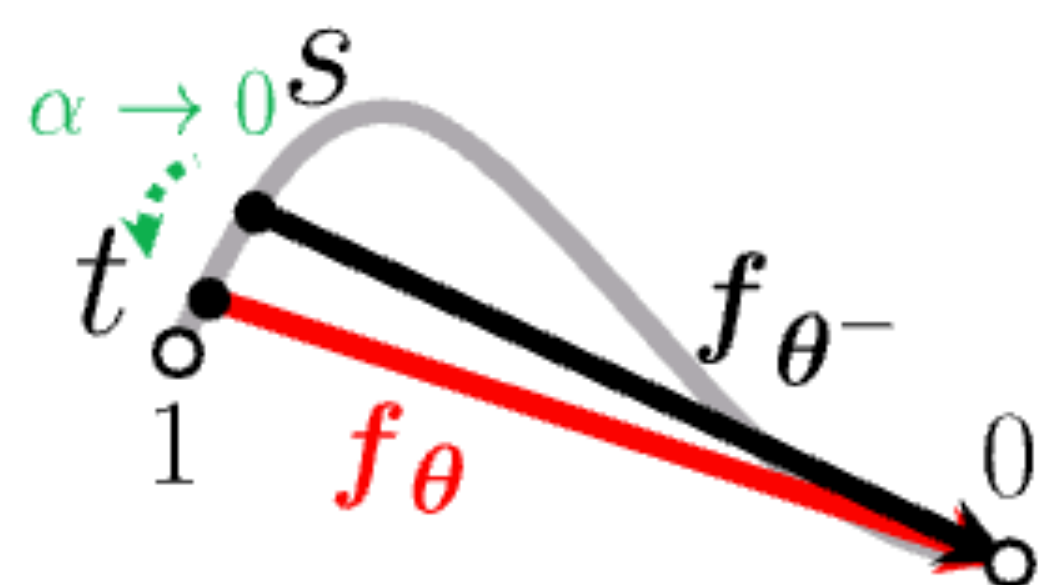
$$\nabla_{\theta} \mathcal{L}_{\theta}^{MF} = \nabla_{\theta} \mathcal{L}_{\theta}^{FlowMaps}$$

Connections

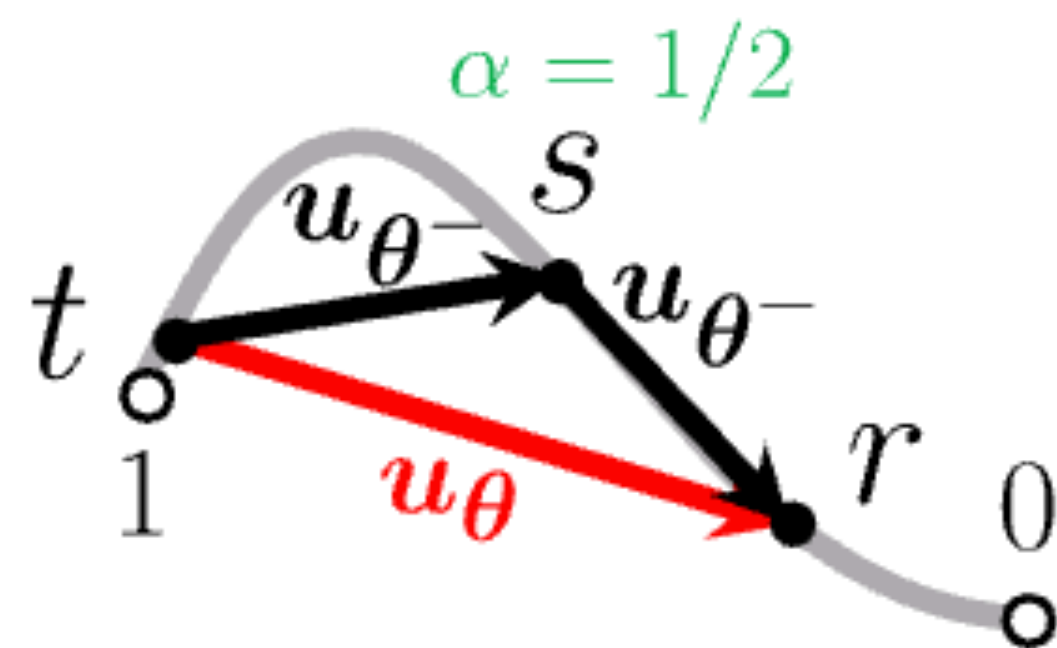
1. Consistency Trajectory Models is a multistep version of Consistency Models
2. Shortcut models is a multistep version of Consistency Models, but in a different parameterization
3. Flow Maps is a continuous version of CTM
4. MeanFlows
 1. Continuous version of Shortcut models
 2. Continuous version of CTM, but in a different parameterization
 3. Equivalent to Flow Maps in terms of gradients



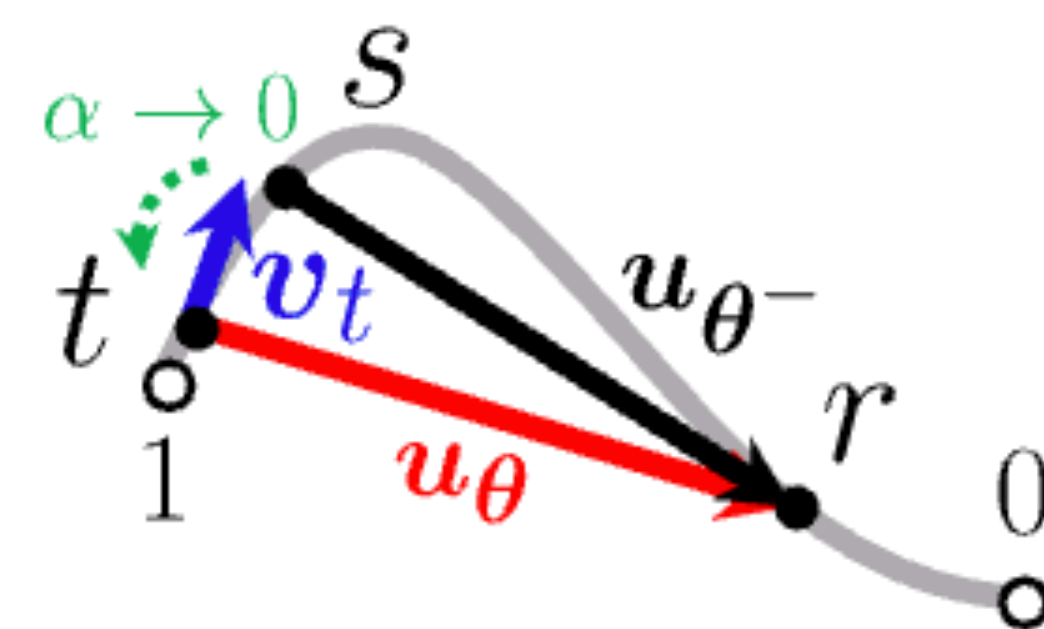
(a) Discrete CT



(b) Continuous CT



(c) Shortcut Model



(d) MeanFlow